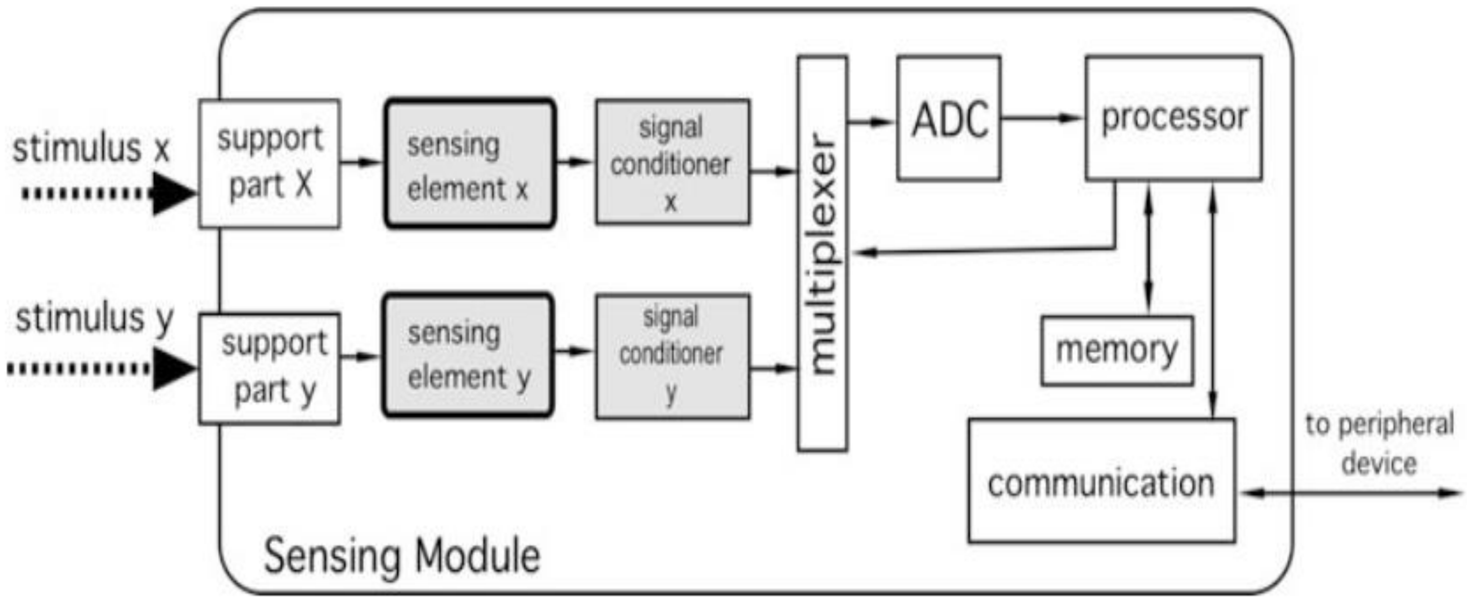
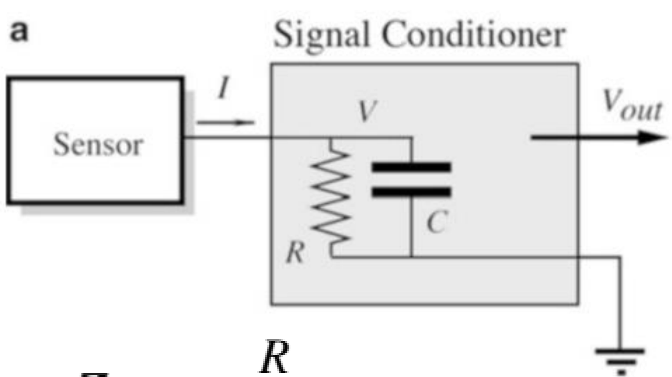
# Interface Electronic Circuits

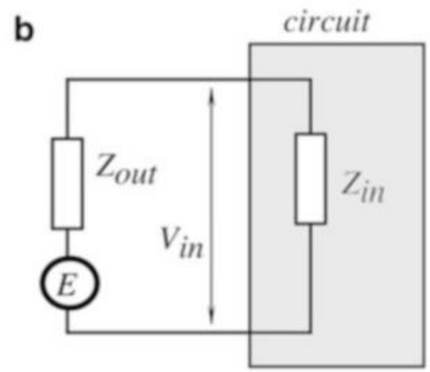
# Signal Conditioner Circuit

Measurement systems block diagram



# Circuit Input impedance




$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}},$$

$$\mathbf{Z} = \frac{R}{1 + j\omega RC},$$

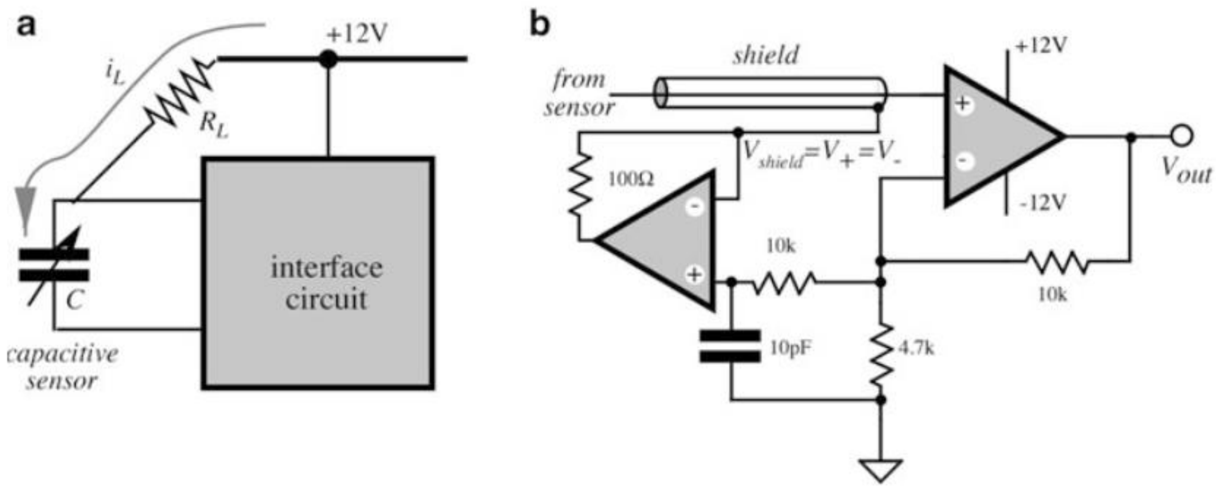



**Fig. 6.4** Circuit board leakage affects input stage (**a**); driven shield of input stage (**b**)

# Current to Voltage

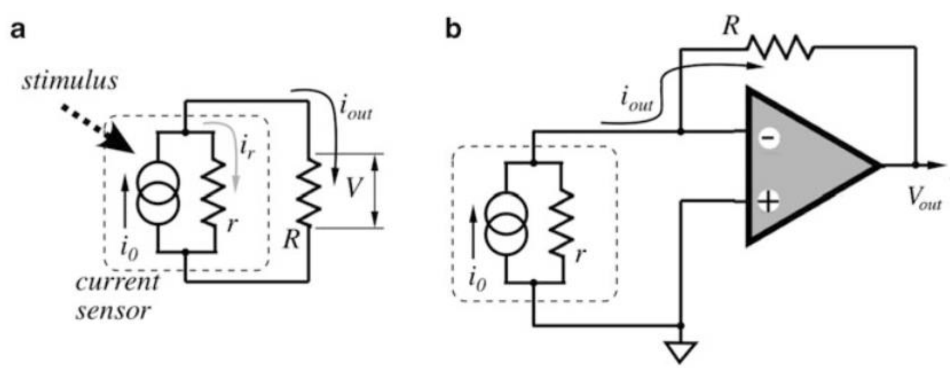


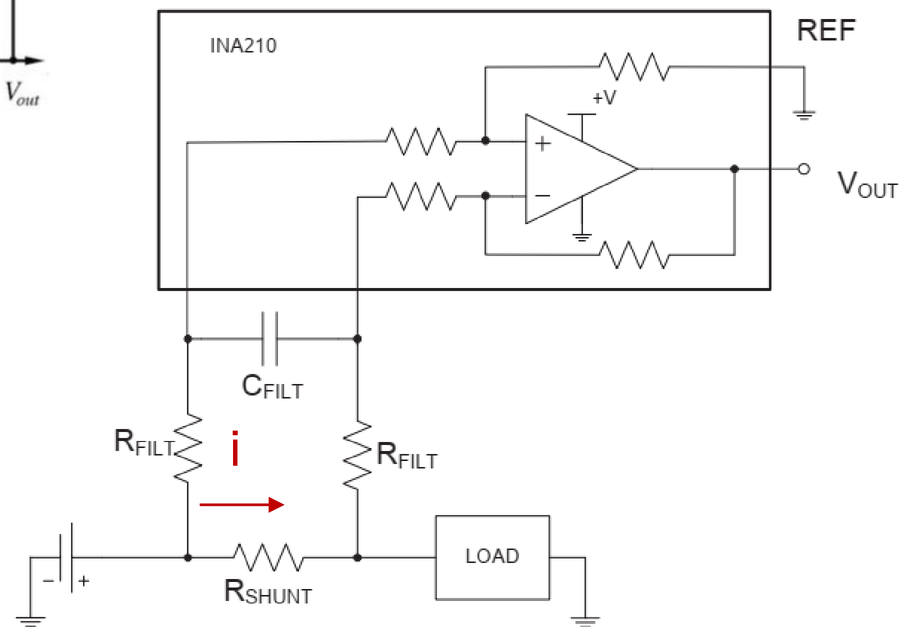

**Fig. 6.7** Current-to-voltage converters

$$V_{\text{out}} = -iR$$

Shunt Resistor

$$G_{\text{INA210}} = 100 \cdot \frac{R_{\text{INA210}}}{R_{\text{INA210}} + R_{\text{FILT}}}$$

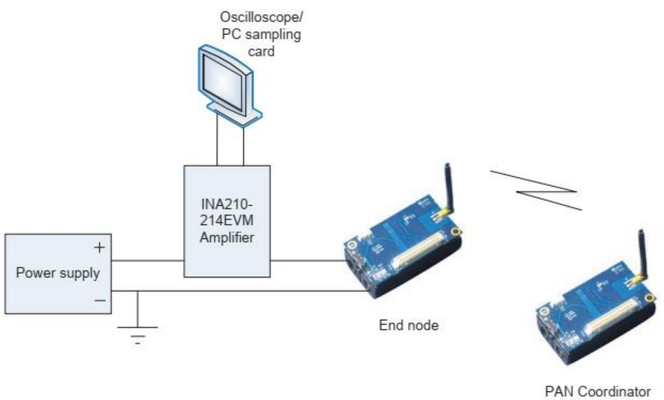
DOI: 10.1109/TSP.2011.6043770 · Source: DBLP

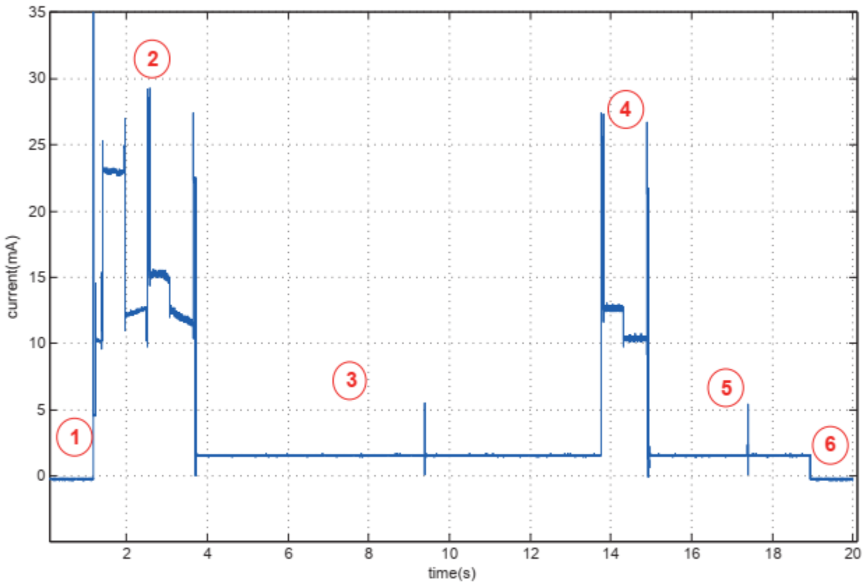



$$f_c = \frac{1}{2\pi \cdot (2R_{FILT}) \cdot C_{FILT}} \tag{3}$$

If we select a value 220 nF for $C_{\text{FILT}}$ we get a cutoff frequency 7.7 kHz, which allows us to use a sampling frequency not lower than 15.4 kHz in accordance with the Nyquist theorem.

# Current to Voltage




**Phase 1:** the node is turned off and the consumption is certainly 0.
**Phase 2:** Turning on the sensor node performs association and binding phase exchanging information with PAN Coordinator and initial packet transmission (phase 2).
**Phase 3:** node waits in a sleep mode (phase 3)
**Phase 4:** draining minimum of energy until periodic packet transmission takes place in phase 4.
**Phase 5** represents waiting for next periodic packet transmission in the sleep mode, however, the end node is turned off (phase 6).

# Light to Voltage

Light-to-voltage converters are required for converting output signals from photosensors to voltage

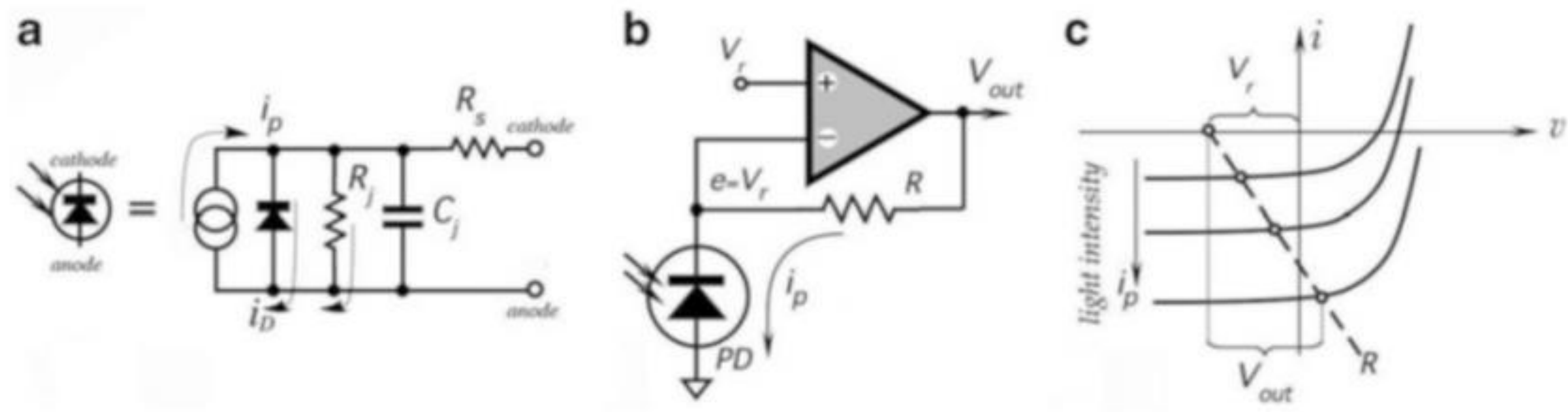



**Fig. 6.8** Equivalent circuit of photodiode (**a**). Reverse-biased photodiode with current-to-voltage converter (**b**). Load diagram of circuit (**c**)

# Light to Voltage

## Photodiode Conditional circuit

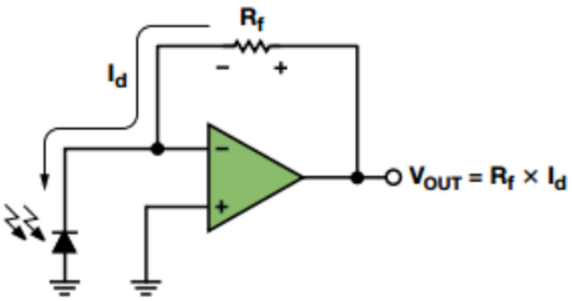



*Figure 1. Simple Transimpedance Amplifier Circuit*

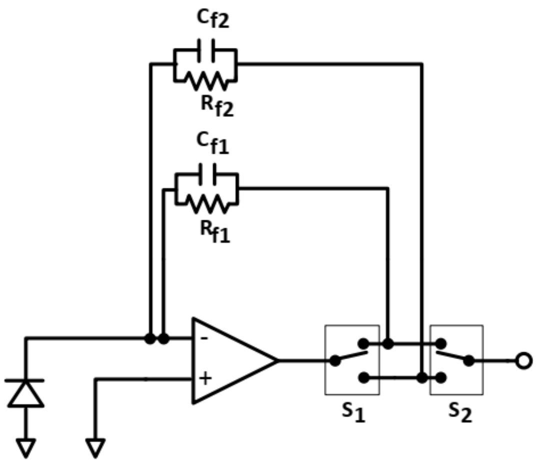



*Figure 8. Using Two Sets of Switches Reduces Errors Due to Additional Resistance Inside the Loop*

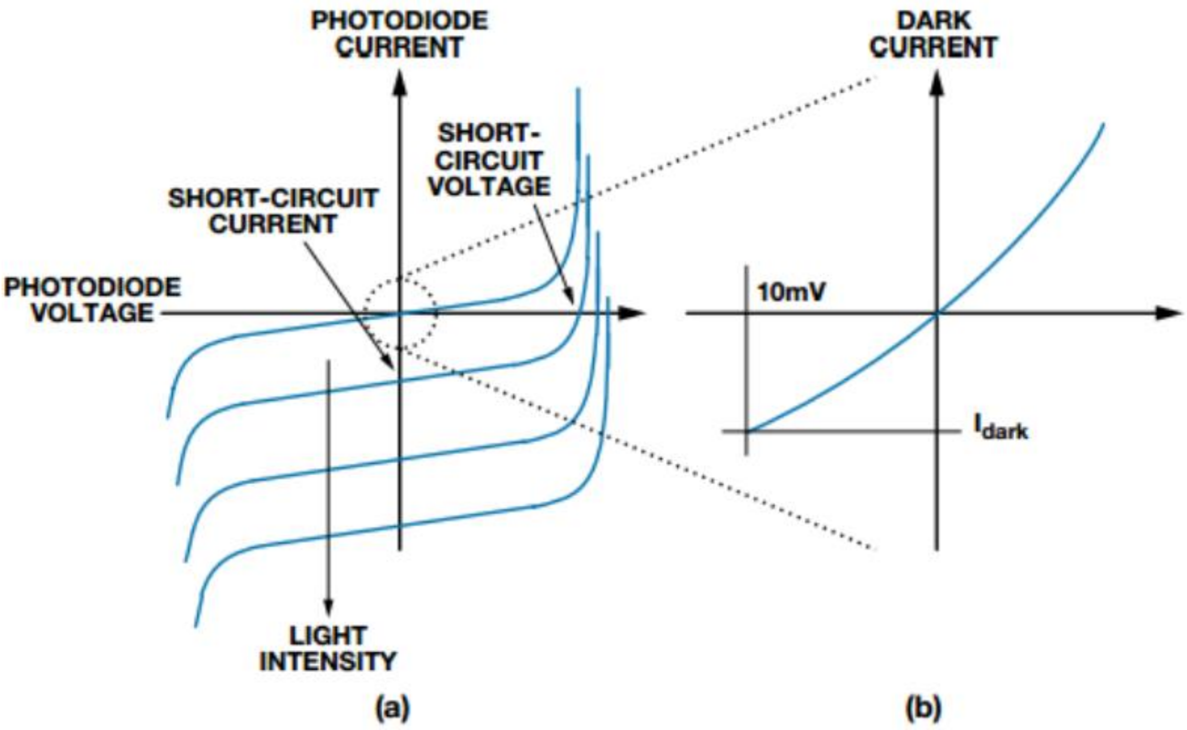



*Figure 2. Typical Photodiode Transfer Function*

# Capacitance to Voltage Converter

Design of Capacitance to Voltage Converter for Capacitive Sensor Transducer

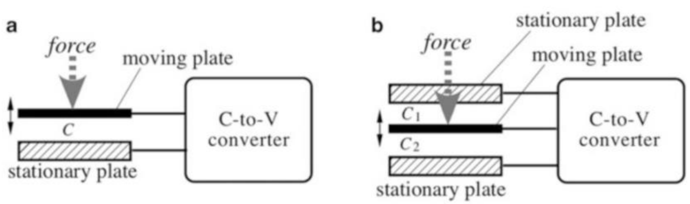



**Fig. 6.10** Asymmetrical (**a**) and symmetrical (**b**) capacitive displacement sensor

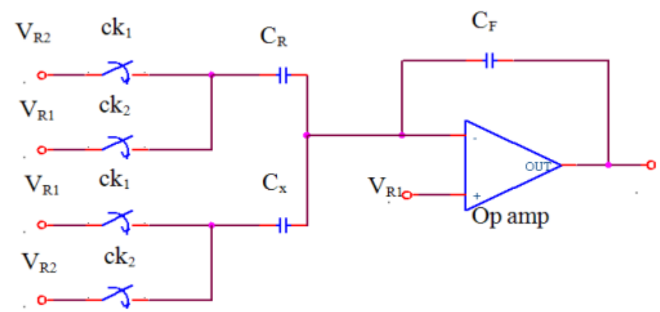



Fig. 1: Capacitance to Voltage Converter (CVC)

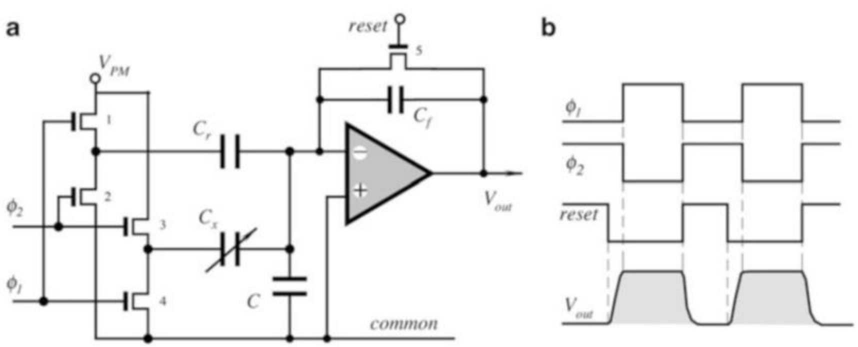



**Fig. 6.11** Simplified schematic (**a**) and timing diagrams (**b**) of capacitance-to-voltage converter using switched-capacitors technique

$$V_o = \frac{C_x - C_R}{C_F}(V_{R2} - V_{R1}) \times \left( \frac{1}{1 + \frac{1}{A}\left(\frac{C_F + C_R + C_x}{C_F}\right)} \right)$$

$$= \frac{C_x - C_R}{C_F}(V_{R2} - V_{R1}) \times \left( 1 - \frac{1}{A\left(\frac{C_F}{C_F + C_R + C}\right)} \right) \quad (5)$$

# Capacitance to Voltage Converter

OPA: bandwidth gain is approximately 1.8 MHz.

The detection capacitance $C_x$ is varied from 20- 1700 fF

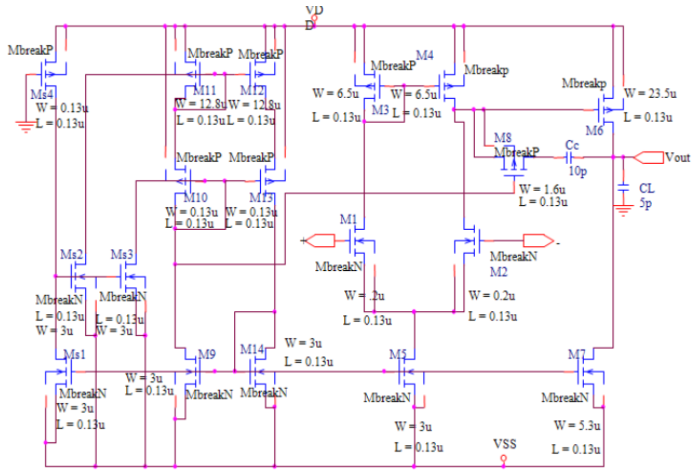



Fig. 2: Schematic for Op amp

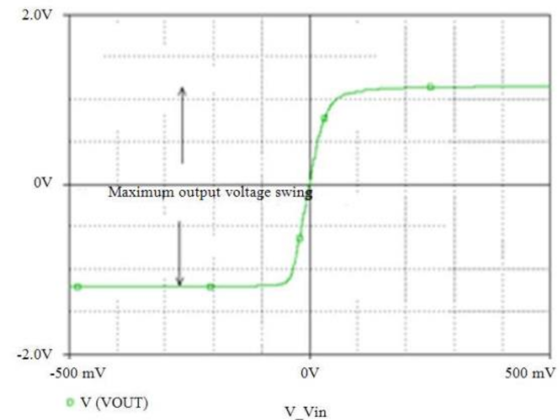
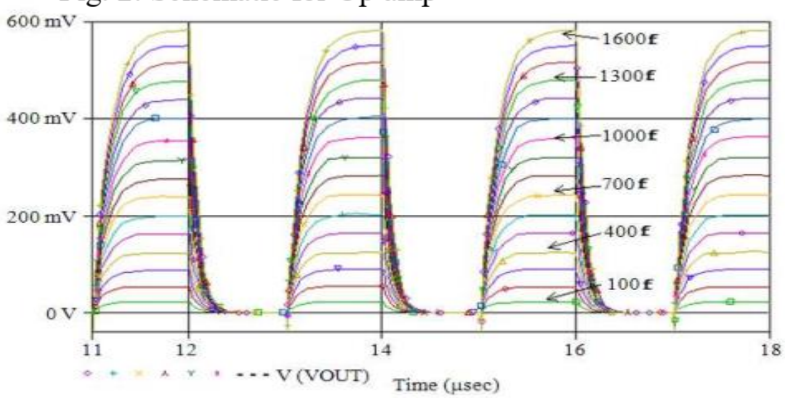



Fig. 3: Voltage transfer characteristics of the Op amp



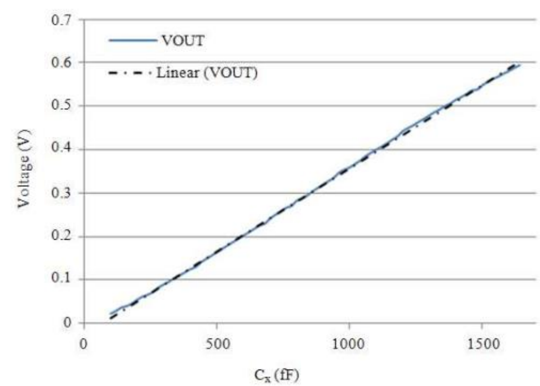




Fig. 6: Peak output voltage with the variation of $C_x$

# Excitation Circuit

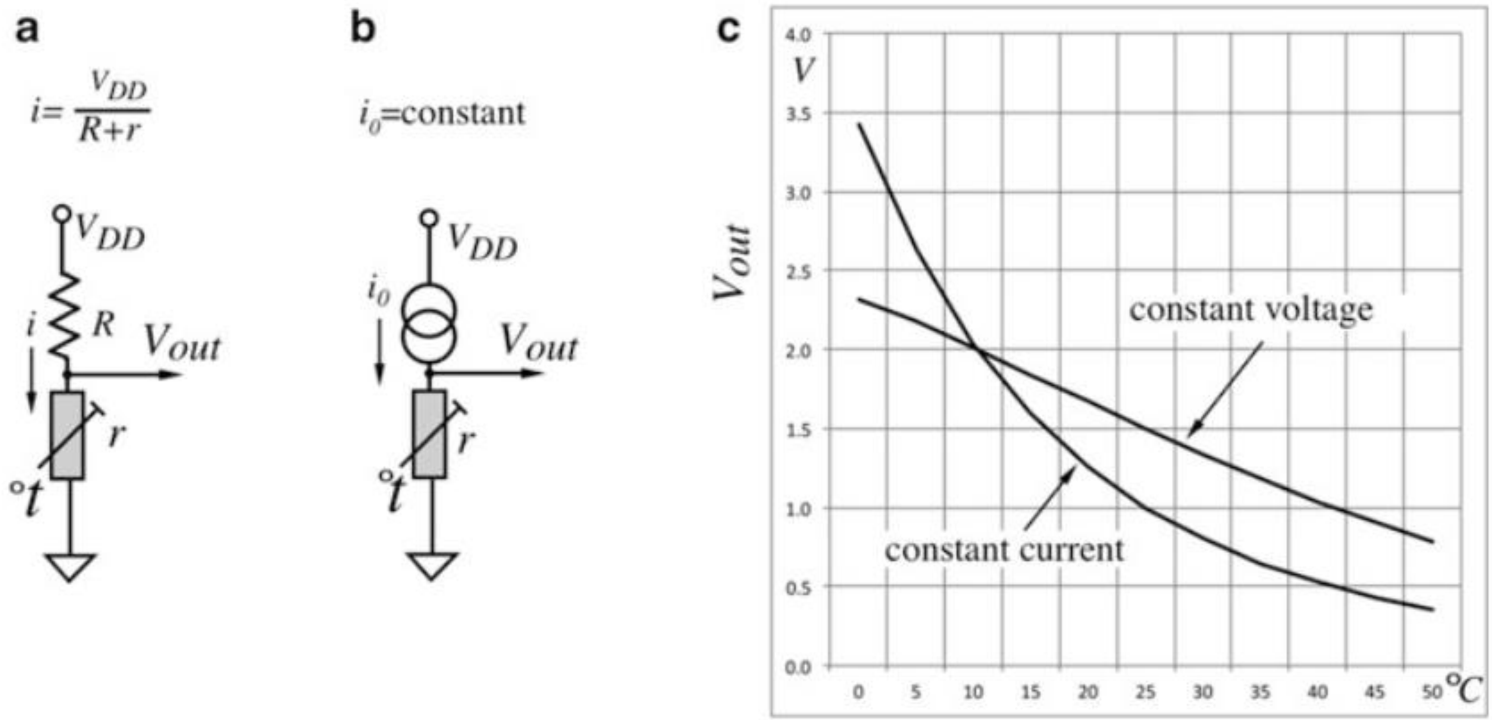



**Fig. 6.20** Generating of excitation currents by a constant voltage source $V_{DD}$ (**a**) and by constant current source $i_0$ (**b**); voltages across thermistor $r$ at two different excitation currents as functions of temperature (**c**)

# Excitation Circuit- Current Source

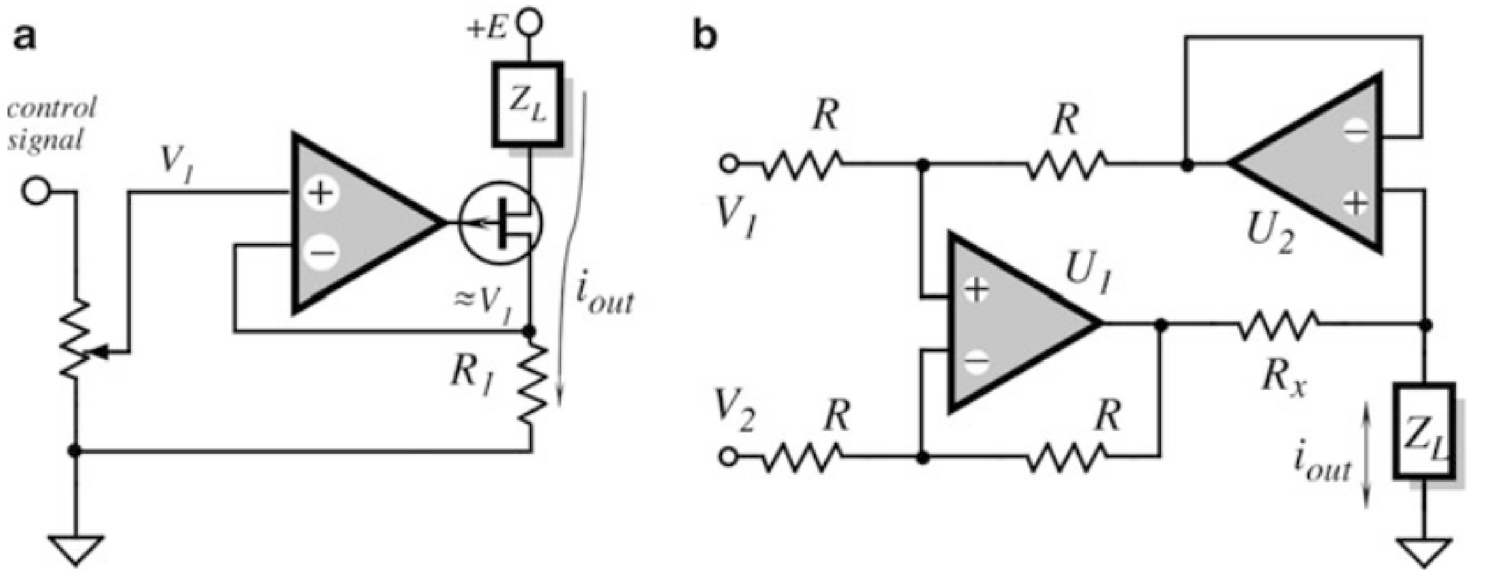


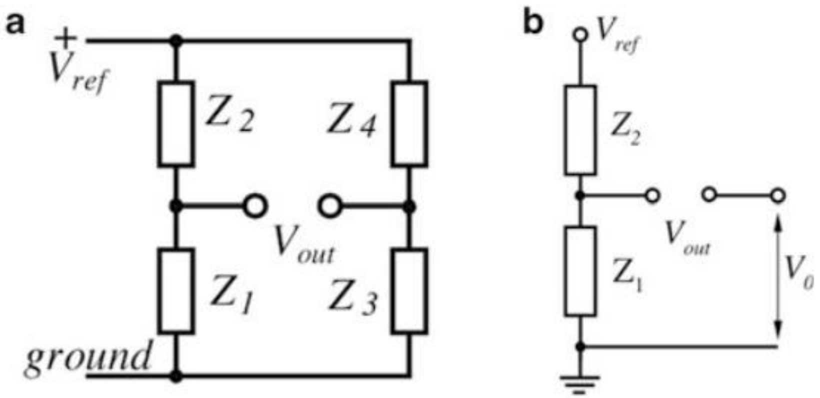

**Fig. 6.21** Current sink with JFET transistor (**a**) and Howland current pump (**b**)

$$i_{\mathrm{L}} = \frac{(V_1 - V_2)}{R_{\mathrm{s}}}.$$

# Wheatstone Bridge Circuit

- Wheatstone Bridge

**Fig. 6.16** General circuit of Wheatstone bridge (**a**) and half-bridge (**b**)




$$V_{\text{out}} = \left(\frac{\mathbf{Z_1}}{\mathbf{Z_1}+\mathbf{Z_2}} - \frac{\mathbf{Z_3}}{\mathbf{Z_3}+\mathbf{Z_4}}\right)V_{\text{ref}} = V_{\text{ref}}\left[\left(1+\frac{\mathbf{Z_2}}{\mathbf{Z_1}}\right)^{-1} - \left(1+\frac{\mathbf{Z_4}}{\mathbf{Z_3}}\right)^{-1}\right]$$

$$\frac{\partial V_{\text{out}}}{\partial \mathbf{Z_1}} = \frac{\mathbf{Z_2}}{(\mathbf{Z_1}+\mathbf{Z_2})^2}V_{\text{ref}}$$

$$\frac{\partial V_{\text{out}}}{\partial \mathbf{Z_2}} = -\frac{\mathbf{Z_1}}{(\mathbf{Z_1}+\mathbf{Z_2})^2}V_{\text{ref}}$$

$$\frac{\partial V_{\text{out}}}{\partial \mathbf{Z_3}} = -\frac{\mathbf{Z_4}}{(\mathbf{Z_3}+\mathbf{Z_4})^2}V_{\text{ref}}$$

$$\frac{\partial V_{\text{out}}}{\partial \mathbf{Z_4}} = \frac{\mathbf{Z_3}}{(\mathbf{Z_3}+\mathbf{Z_4})^2}V_{\text{ref}}$$

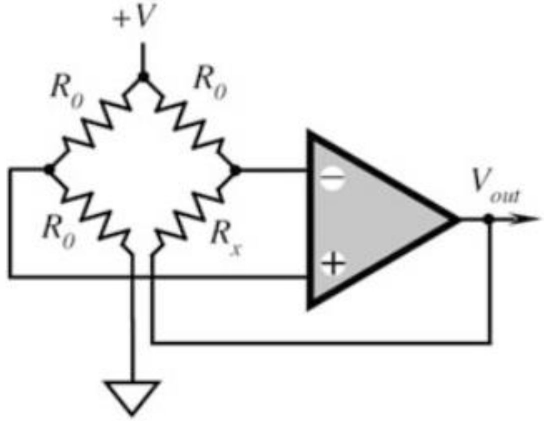
If (Z1= Z2) $\rightarrow$ $$\frac{\delta V_{\text{out}}}{V_{\text{ref}}} = \frac{\delta \mathbf{Z_1}}{4\mathbf{Z_1}}$$

# Wheatstone Bridge Circuit

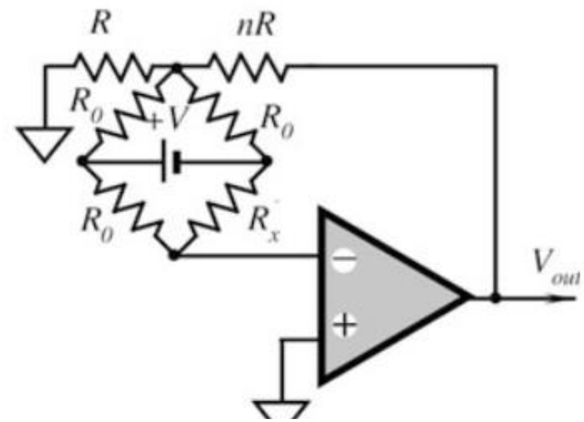
**a**

$$R_x \approx R_o(1+\alpha),$$

$$V_{\text{out}} = \frac{V}{2}(1-\alpha).$$



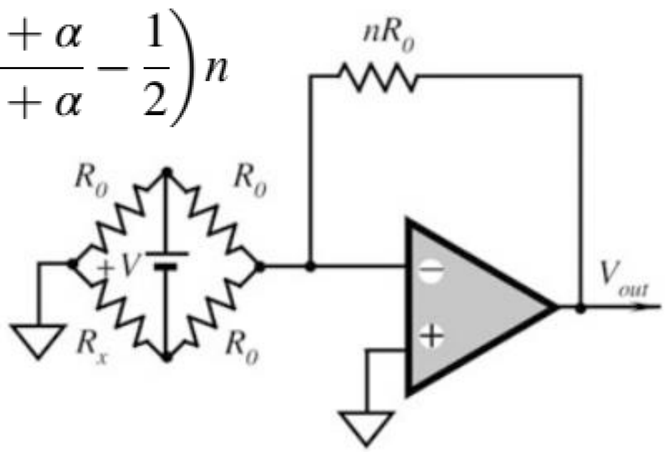

**b**



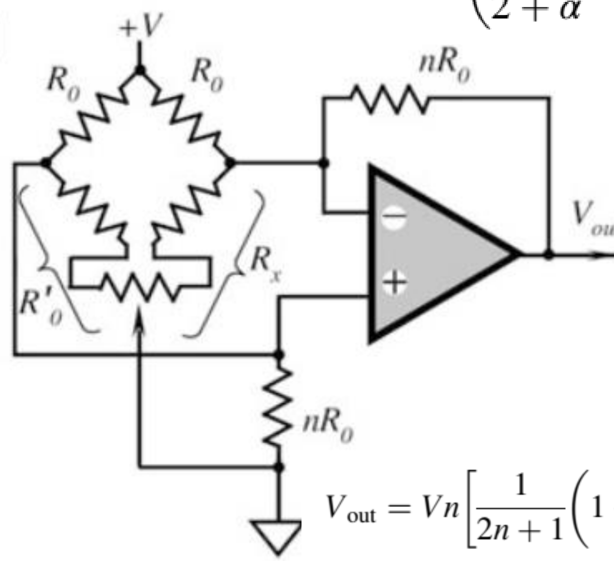

$$V_{\text{out}} = V\left(\frac{1+\alpha}{2+\alpha} - \frac{1}{2}\right)(n+1)$$

**c**

$$V_{\text{out}} = V\left(\frac{1+\alpha}{2+\alpha} - \frac{1}{2}\right)n$$




**d**




$$V_{\text{out}} = Vn\left[\frac{1}{2n+1}\left(1+n\frac{2+\alpha}{1+\alpha}\right) - 1\right]$$

# ADC Interface

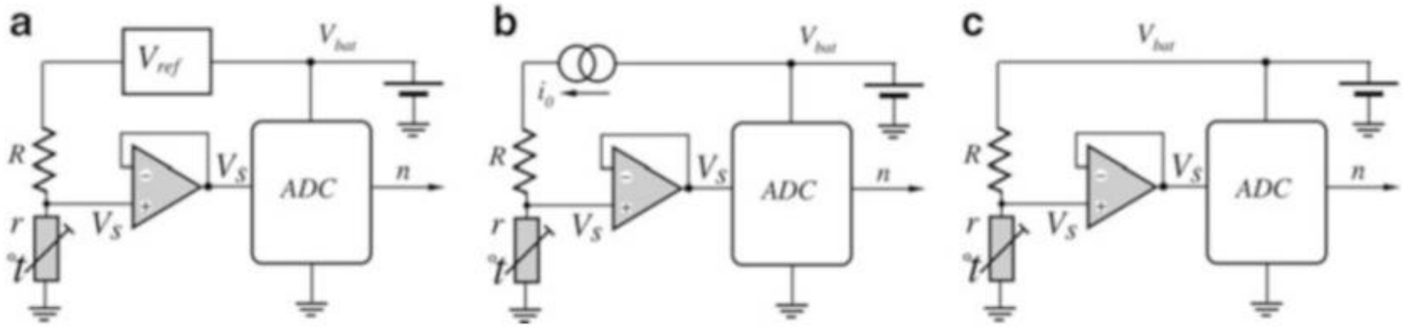



**Fig. 6.33** Powering the sensor circuit with constant voltage (**a**), constant current (**b**), and ratiometric voltage (**c**)

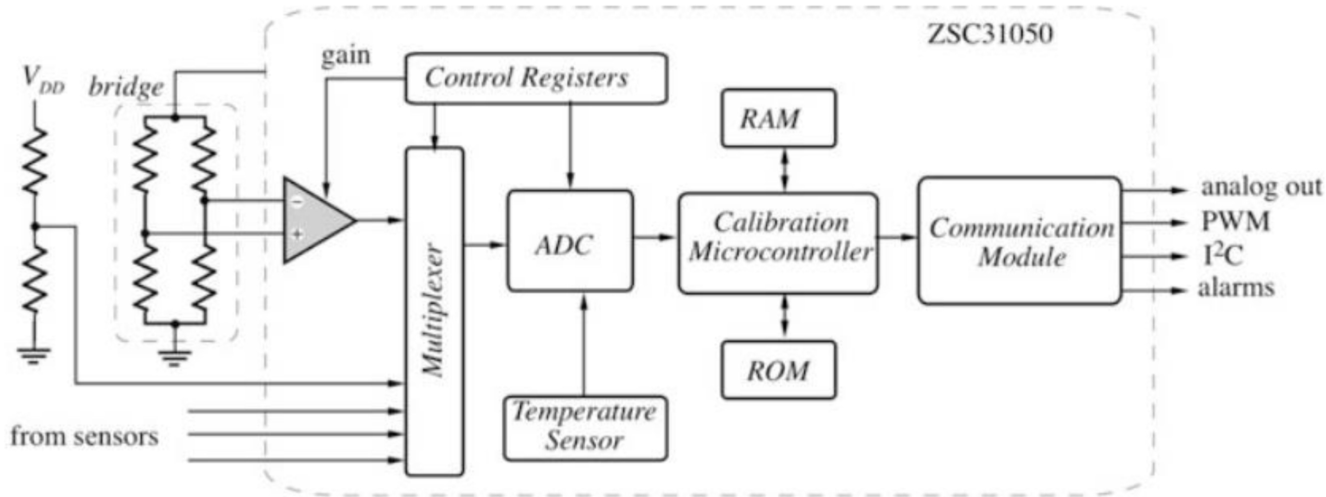
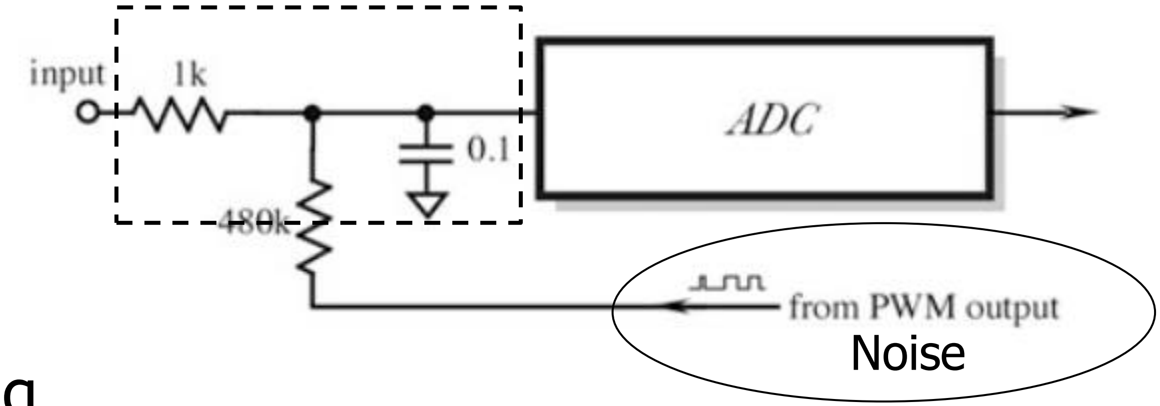



**Fig. 6.34** Integrated ZMDI Signal Conditioner

# ADC Interface

- ADC input

Analog Low-pass filter




- ADC Sampling

Digital Filter:
- Average Number of Samples per one point
- Design a digital Low-pass filter with code
- Kalman Filter

# Noise in Sensor and Circuit

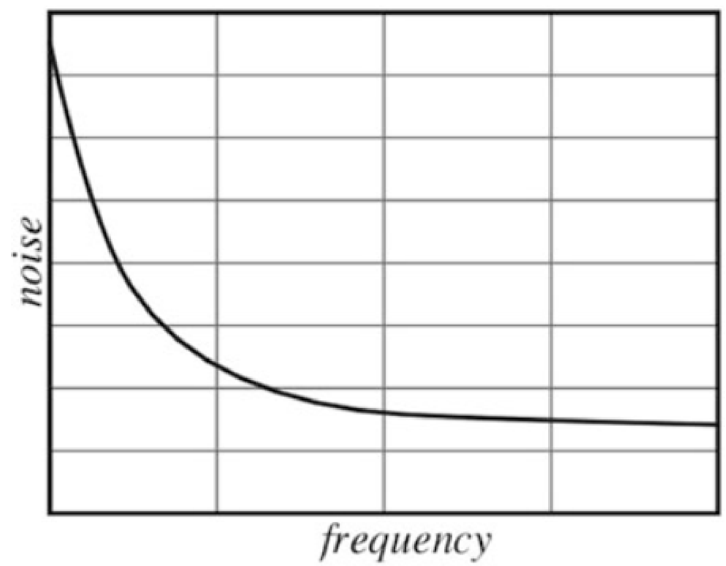
## Inherent Noise

The movement of carriers are temperature related and noise power, in turn, is also temperature related. In a resistor, these thermal motions cause Johnson noise. The mean-square value of noise voltage (which is representative of noise power) can be calculated from:

$$\overline{e_n^2} = 4kTR\Delta f \quad [\mathrm{V^2/Hz}]$$

$$\downarrow$$

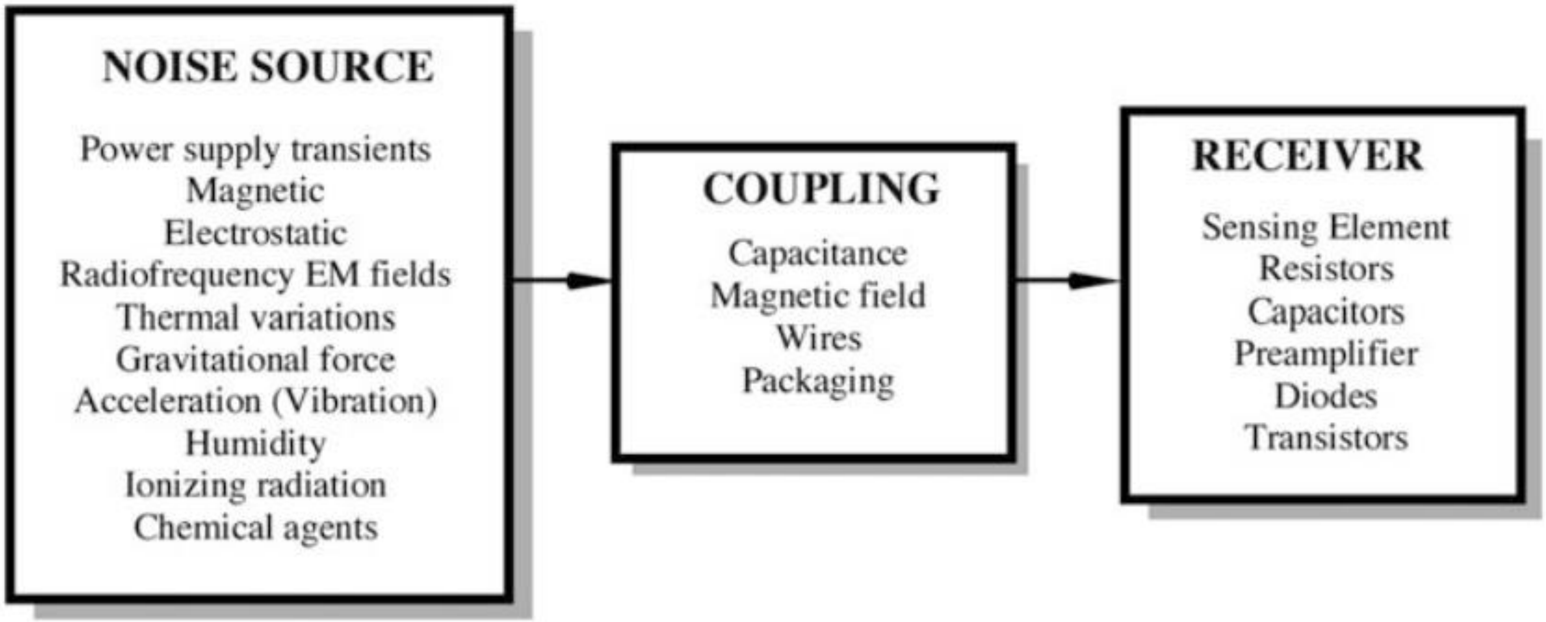
$$\overline{e_n} \approx 0.13\sqrt{R \cdot \Delta f} \quad \text{in nV.}$$




Spectral distribution of 1/f "pink" noise

# Noise in Sensor and Circuit

- Transmitted Noise

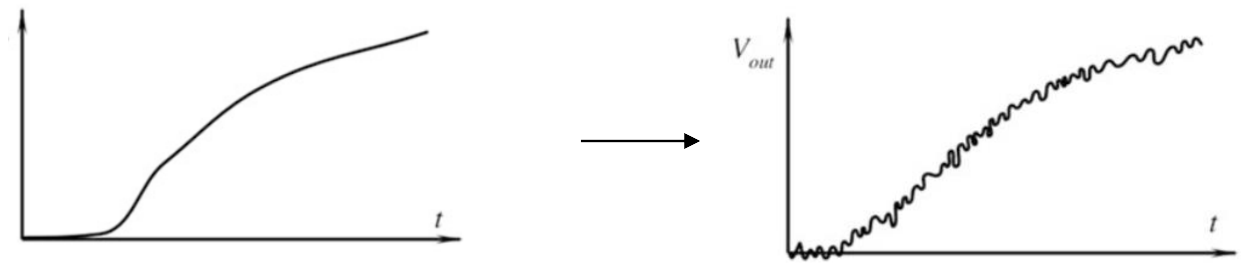



Sources and coupling of transmitted noise

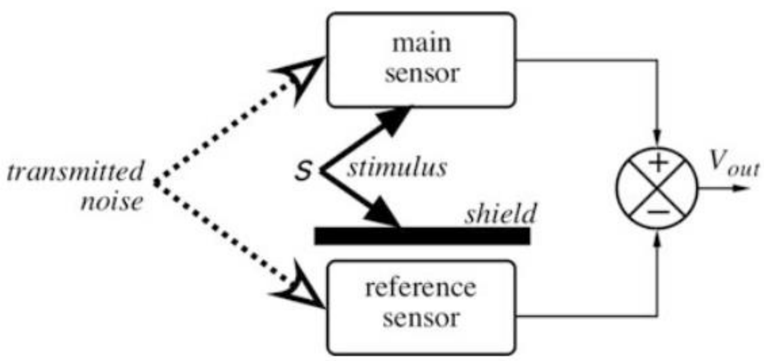
# Noise in Sensor and Circuit

- Additive Noise

$$V_{\text{out}} = V_{\text{s}} + e_{\text{n}}.$$




- Shield prevents stimulus s from reaching reference sensor




$$CMMR = 0.5\frac{S_1 + S_0}{S_1 - S_0}$$

# Noise in Sensor and Circuit

- Improve noise-Electrical Shield

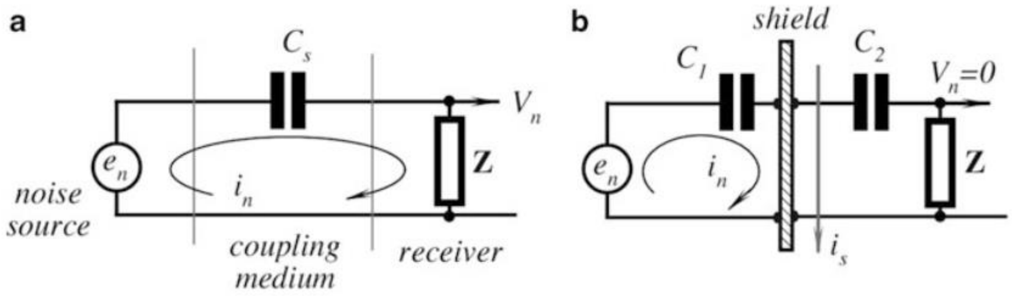
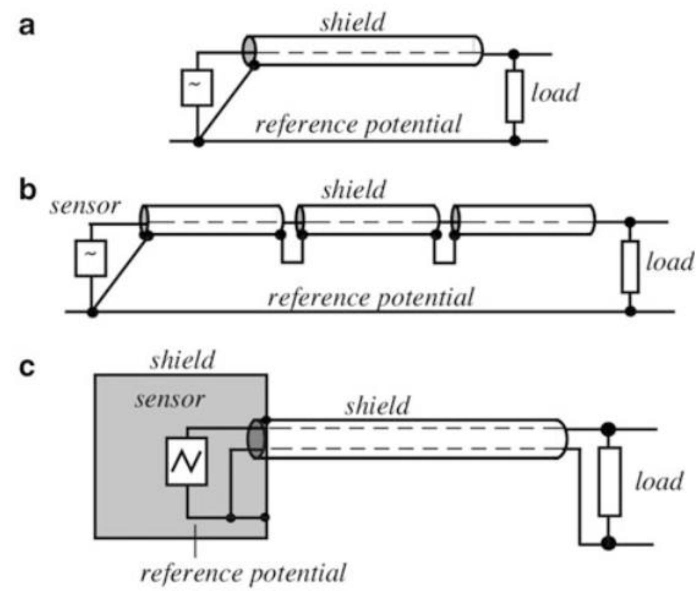
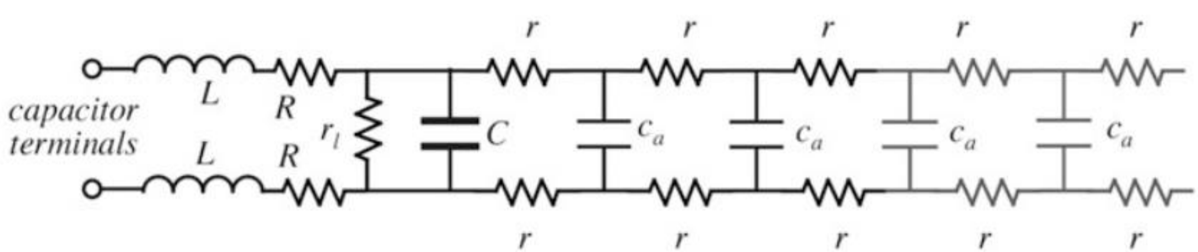



**Fig. 6.43** Capacitive coupling (**a**) and electric shield (**b**)




- Bypass Capacitors



# Noise in Sensor and Circuit

- Seebeck Noise

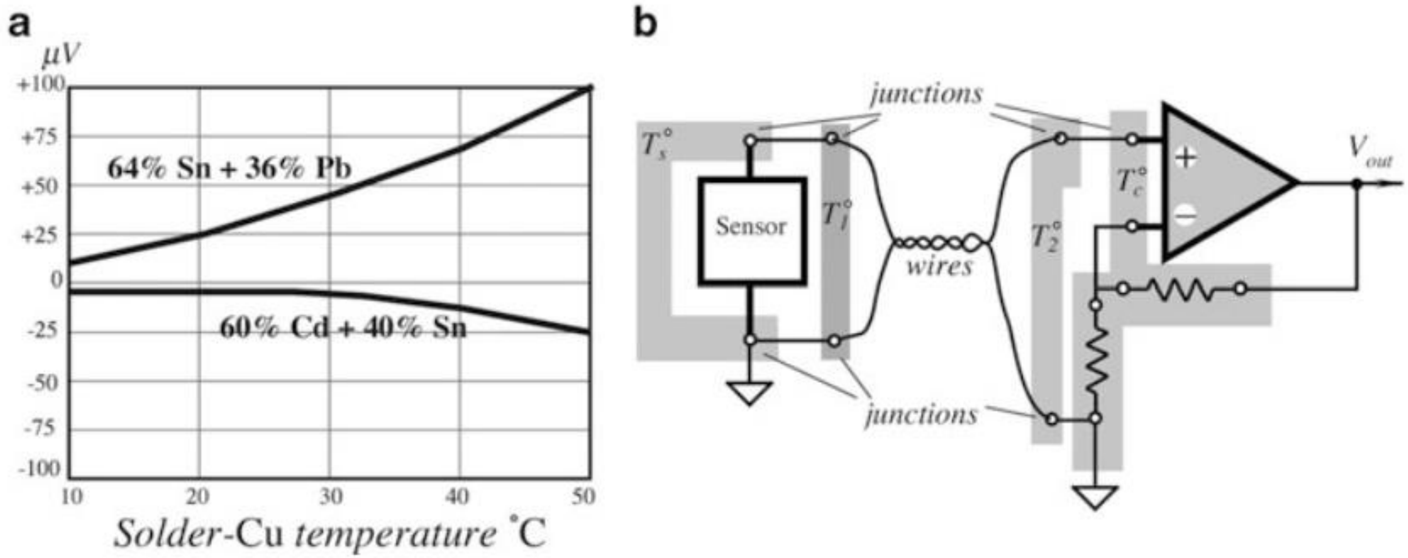



**Fig. 6.51** Seebeck *e.m.f.* developed by solder-copper joints (**a**) (adapted from [14]); maintaining joints at the same temperature reduces Seebeck noise (**b**)